

Adjusting Level of Abstraction for Stylized Image Composition

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ABSTRACT

Recently, many image stylization techniques have been proposed to convert photographs into illustrations, paintings, cartoons, etc. Artists sometimes use these techniques to efficiently generate background images for their own illustrations of foreground objects, such as cartoon characters. However, this approach often produces unnatural results because the level of abstraction between the foreground and the background images are different, resulting in inconsistent atmosphere between them. In this paper, we address this problem and propose a method for adjusting the abstraction level of the stylized background images. The effectiveness of our method is demonstrated by several examples with different levels of abstraction.

KEYWORDS

illustration, style transformation, entropy, semantic segmentation

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1 INTRODUCTION

Automatic image stylization has a long history and many methods have been proposed. They are used for different purposes including artwork, movies, video games, etc. Recently, a machine learning approach has made a remarkable progress in style transformation. One of the latest models, Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization (AdaIN), can produce stylized images similar to a given exemplar.[Huang and Belongie 2017] AdaIN takes a content image and a style image as input and transfers the style of the style image into the content image.

Our goal is using AdaIN for creating stylized background images from a photograph for a user-specified illustration of a foreground character. This is achieved by using the photograph as the content image and the character illustration as the style image. However, this straightforward approach often fails to produce convincing results; the atmosphere of the background image is inconsistent with that of the foreground illustration. The inconsistency is caused by the difference in levels of abstraction between them. In this study, we propose a method to measure the levels of abstraction in the

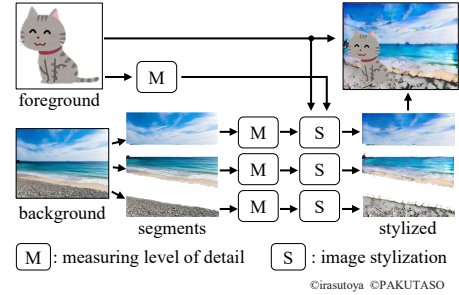


Figure 1: Overview of our method.

foreground illustration and the background photograph, respectively, and adjust the level of abstraction of the stylized background image to match that of the foreground illustration.

2 OUR METHOD

We assume that the level of abstraction of an image depends on the level of details in the image. We want to balance the levels of abstraction between the foreground and the background by removing the details from the input image for the background. Since the level of detail in an image may change from pixel to pixel, so we first apply a segmentation technique to the input image and adjust the abstraction level for each segment.

Fig. 1 shows an overview of our method. Our method uses two input images shown on the left; one represents an illustration of a foreground character and the other a photograph for creating the background image. The method first measures the level of detail of the foreground image. The background photograph is then decomposed into segments so that the level of details in each segment seems uniform. For each segment, we measure the level of detail and AdaIN is used to stylize it. We adjust a parameter of AdaIN system to make the level of detail of the stylized segment similar to that of the foreground image. Finally, the segments are combined to generate the stylized background image.

In the following, we first describe our method for measuring the level of detail in Section 2.1. Section 2.2 then describes the method to create the stylized background image with the adjusted level of details.

2.1 Measuring Level of Details

We measure the level of details of the foreground image by the entropy of the color distribution and the amount of the edges in the image. The measured level of details is used to determine the parameters for stylization of the background image.

The entropy is calculated from the hue histogram of the image. We calculate the hue value for each pixel quantized into 256 bins.

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Figure 3: Examples of composite images with three characters with different levels of details.

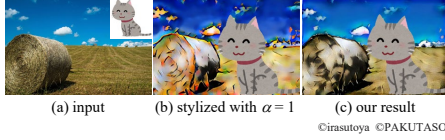


Figure 2: An example of stylization.

The entropy H is then expressed by the following equation.

$$H = \sum_{i=1}^{256} (-p_i \times \log_2 p_i), \quad (1)$$

where p_i is the number of pixels in i th bin. The entropy becomes larger when the number of the colors increases.

The entropy alone is not sufficient to measure the level of details. Even though the entropy is small (i.e., similar colors are used), the image with detailed edges looks complicated. Therefore, we additionally use the amount of edges in the image. We use the Canny edge detector to find the pixels corresponding to the edges and we count the number of the edge pixels. The number is then normalized by the total number of the pixels of the image. Note that, for the foreground image, we use only the pixels corresponding to the foreground object in the image.

Finally, the level of details of the image is defined as the weighted sum of the entropy and the edge amount.

2.2 Stylization of Background

We use AdaIN for stylization of the background image provided by the user. AdaIN can transfer the style of the foreground image into the background image. AdaIN has a single parameter α that allows the user to control the degree of the stylization. The background image is the most stylized when $\alpha = 1$. We adjust this parameter so that the level of the detail of the stylized background becomes same as that of the foreground image. However, the level of details in the background image could change from pixel to pixel. If we use a single value for α for the entire image, some regions in the background may be too abstracted. We thus decompose the background image into segments with similar levels of details. Then, each segment is stylized with an appropriate parameter value.

In our current implementation, we use a simple segmentation method, the Watershed algorithm [Beucher and Meyer 1993]. We manually specify several pixels as seed points for the segmentation. Next, the stylization parameter (α) is determined for each segment with the following heuristic method. For segment i , its parameter

α_i is determined by the following equation.

$$\alpha_i = 1.0 - q_f + D_i, \quad (2)$$

where q_f is the level of detail computed from the foreground image. D_i is determined by the following equation.

$$D_i = \begin{cases} -0.2 & (q_i \leq q_f) \\ 0.2 & (q_i > q_f) \end{cases}, \quad (3)$$

q_i is the level of detail computed for segment i .

The above method is derived from our experiments, but automatic segmentation and the optimization of the parameter values are important future work.

3 EXPERIMENTAL RESULTS

Fig. 2 shows an example of stylization. Fig. 2(a) shows the input photograph used for the background. The foreground illustration is shown in the inset at the top right corner. The input photograph is segmented into two (the sky and the ground). Fig. 2(b) is created with $\alpha = 1$ and without segmentation. Fig. 2(c) show our result. This example demonstrates that the parameter is successfully adjusted to produce the stylized background with similar details to the foreground image.

Fig. 3 shows examples with three different foreground illustrations. The input photograph for the background is the same as that of Fig. 2. The parameter α is automatically adjusted by our method. This example demonstrates that our method successfully generates the stylized background with consistent atmosphere with the foreground illustration. More examples can be found in the supplemental material.

4 CONCLUSIONS

In this paper, we have proposed the method for stylizing background images based on the level of details of the foreground illustrations. Our method automatically determines the stylization parameter so that the levels of details between them are balanced, resulting in natural composition. For our future work, we want to achieve more natural composition by removing the discrepancy of lighting between the foreground and the background images.

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